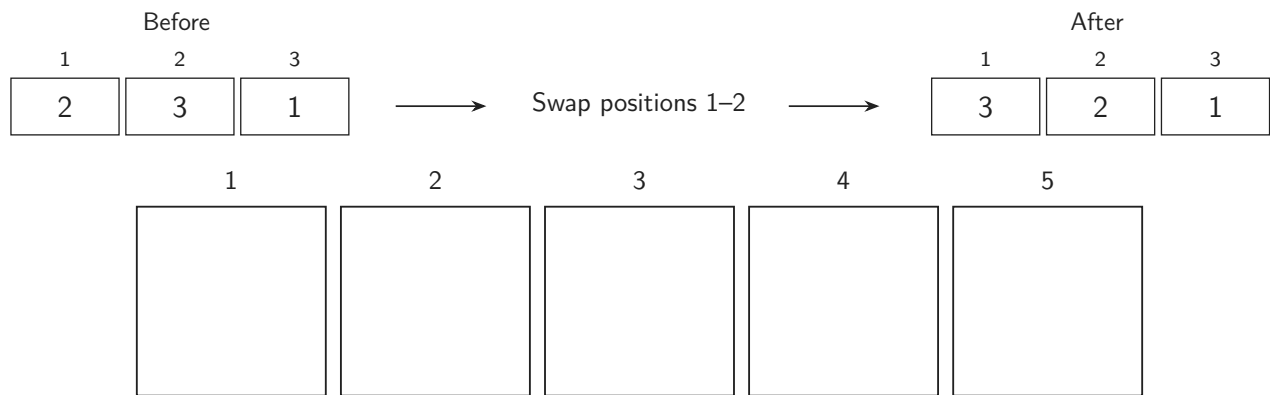


Problem 1: Swap two cards in neighboring positions. Put each row into the order 1, 2, 3, 4, 5 using as few swaps as you can. Record the positions you swap, such as 2–3. Arrange cards 1–5 into a row that needs as many swaps as possible.



2	1	3	5	4
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3	1	4	2	5
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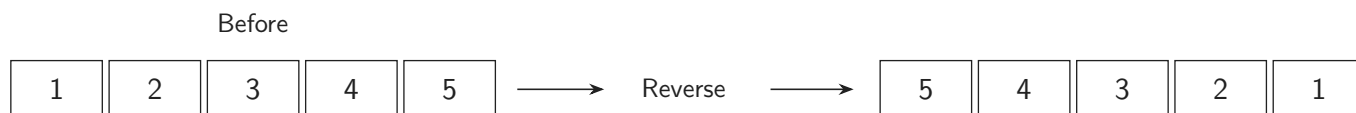
5	3	4	2	1
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My hardest row:

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1	2	3	4	5	6
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Problem 2: Start with 1, 2, 3, 4. A move takes the last card to the front, or reverses the whole row. Make each target, or cross it out if you think it cannot be made. Find every order these moves can make.



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Problem 3: Draw one arrow from each input slot. Each output slot must get exactly one arrow. Move each card along its arrow. Put the output cards straight back into the same input positions and use the same arrows again. Starting with the cards in number order, make each target after two passes. Use cards on the table to test your machines. Cross out a target if you think it is impossible.

